

- [34] Li, Y., Fan, H., Hu, R., Feichtenhofer, C. & He, K. Scaling language-image pre-training via masking. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 23390–23400 (2023).
- [35] Wu, P. *et al.* Masked trajectory models for prediction, representation, and control. *arXiv preprint arXiv:2305.02968* (2023).

Appendix

Model variations to avoid overfitting. The same experiments were performed by incorporating dropout layers and random masks into the Transformer model. Dropout [33] is a regularization technique commonly used in neural networks to prevent overfitting and to enhance the model’s capacity for generalization. Specifically, Transformer-based models with a substantial number of parameters have demonstrated better performance by utilizing dropout layers. Thus, despite the relatively modest number of parameters in our Transformer model, we pursued an additional experiment by introducing dropout layers. As in [8], we apply dropout to the output of each sub-layer and the sums of the embeddings and the positional encodings. The base model uses a dropout rate of $P_{drop} = 0.0$.

Figure 4 illustrates the average Dynamic Time Warping (DTW) scores for model outputs across various amounts of training sequences. Transformer models equipped with an appropriately fine-tuned dropout rate ($P_{drop} = 0.01$) demonstrate superior performance compared to models without dropout layers, particularly when dealing with limited training datasets. However, in contrast to recurrent neural network (RNN) models, transformer models have yet to attain a satisfactory level of representation, achieving scores closely aligned with RNNs when training with approximately 25-50 sequences for point attrac-

tors and 10 sequences for cyclic attractors. Also, models with $P_{drop} = 0.1$ and $P_{drop} = 0.3$ exhibit underfitting behavior within this setting. For more detailed scores, see Table 1 and 2.

We also conducted experiments with a combination of random masks. Training with random masks enables different training objectives and combinations, so that more training signals can be extracted from a given trajectory, and thus the model becomes more data efficient [34, 35]. However, the conclusions remained the same, and hence, the results are omitted here.

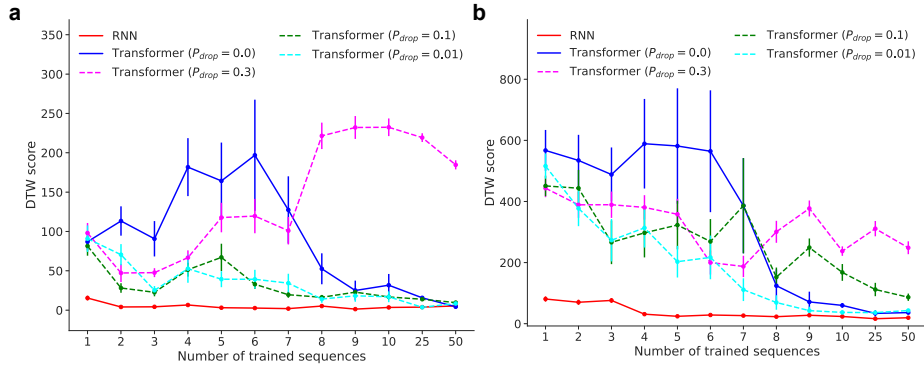


Figure 4: **Model variation study.** **a,b,** The figure presents the average DTW plot along with the standard error for models trained with point and circle attractors, respectively.

Human hand-drawn pattern learning. Given the inherent variability in amplitude, velocity, and shape of human hand-drawn patterns, the generated data is inherently more complex compared to the trajectories employed in the aforementioned study. For this particular investigation, a singular training trajectory was employed (see Figure 5; **a**). This trajectory traced a figure-eight pattern approximately three times, encompassing a total of 560 time steps.

Figure 5; **b** shows that the RNN was able to extract the underlying essential structure, i.e., converging to the figure 8 attractor, while the Transformer model struggles to extract meaningful patterns.

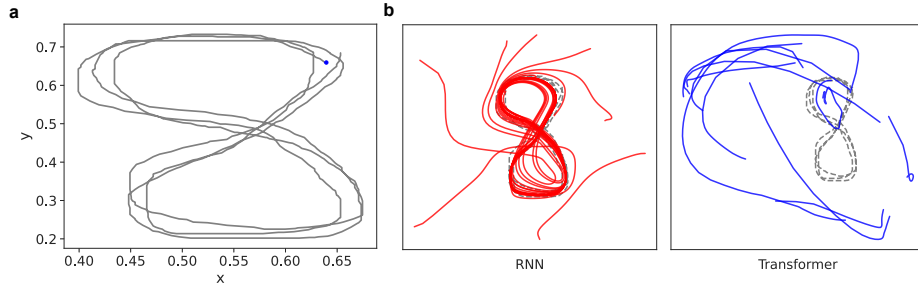


Figure 5: **Result of a human hand-drawn pattern learning.** **a**, Training data. The trajectory is drawn by a human. Blue dot points denote the initial positions. **b**, Model output from ten random initial positions. Black dot lines indicate the trajectory of training data.

Data availability statement. The datasets used during the current study are available from the corresponding author (R.F.) upon request. The codes for this experiment, including the data generation, will also be made available as a GitHub repository.

Author contributions J.T. conceived the concept. R.F. conducted the experiments and wrote the manuscript.

Competing interests declaration The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.